

Given the probability distribution function (PDF) of the original image's Value (V) channel and an RGB image whose V channel has been histogram equalized in the HSV color space, reconstruct the original image.

The histogram equalization was applied only to the V channel, while the Hue (H) and Saturation (S) channels remain unchanged. The inputs are (1) the PDF of the original V channel and (2) the histogram-equalized image in RGB format. The objective is to recover the original RGB image by estimating the original V channel using the provided PDF.

Provided Files

1. boat.jpg – Original image (provided only for comparison with the reconstructed output; do not use it during computation).
 2. boat_hsv_v_pdf – Probability distribution function (PDF) of the original image's V channel.
 3. equalized_image_hsv_v.png – RGB image obtained after histogram equalization of the V channel in the HSV color space. Use this image as the input for reconstruction.
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Output format:

Reconstructed image	PDF of given and reconstructed image	CDF of given and reconstructed image
Equalized Image	PDF of equalized image	CDF of equalized image

